Lower Extremity Venous Doppler

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KEYWORDS

• Color Doppler • Deep vein thrombosis • Ultrasound

Deep venous thrombosis (DVT) can lead to clinically significant complications and mortality if left untreated. These complications include fatal pulmonary embolisms (PEs), recurrent venous thromboembolic events, and postphlebitic syndrome. At least 100,000 deaths per year are attributed to PEs in the United States.[1] Because DVT is the initiating event in PE, the presence or clinical suspicion of a PE should trigger the search for a DVT.

Ultrasound has supplanted conventional venography as the primary imaging modality in screening for and detecting DVTs. Because clinical features often are unreliable and nonspecific in determining the presence of a DVT, imaging of the lower extremity veins remains an important part of the clinical work-up of a suspected PE or DVT. An untreated DVT leaves a patient at a high risk for a thromboembolic event. In one study, more than 50% of thromboembolic events were associated with a DVT.[2] DVT and PE should be treated as part of the same disease process, and any clinical suspicion for a PE or DVT should prompt further investigation.

In addition to reviewing the imaging features of DVT, this article discusses the epidemiology, clinical features, and pathophysiology of lower extremity DVTs.

EPIDEMIOLOGY

The incidence of deep venous thrombosis is 0.5 per 1000 person-years.[3] DVT affects approximately 2 million people, of whom 600,000 suffer from a PE.[4] Venous thromboembolism occurs at an annual incidence of 0.1%.[5,6] Without associated comorbidities, age alone increases the risk for thromboembolic events, escalating from a rate of 0.01% in young adults to approximately 1% in populations over 60 years of age.[5,6] The widespread use of thromboprophylaxis in high-risk hospital inpatients has reduced the incidence of DVT and venous thromboembolism.[7] The use of graded compression elastic stockings and low-dose unfractionated or of low-molecular-weight heparin reduce the risk of developing a DVT by 44% and by 68% to 76%, respectively.[8]

Other risk factors for DVT include smoking, hypertension, varicose veins, cardiac dysfunction, obesity, malignancy, surgery (particularly orthopedic surgery), immobilization, fractures, puerperium, use of oral contraceptives, autoimmune diseases, and thrombophilias.[9–15] Thrombophilias, which may be acquired or inherited; are conditions that favor but do not incite the formation of a thrombus.[9]

CLINICAL MANIFESTATIONS

Patients who have a DVT may be asymptomatic or present with pain, swelling, and lower extremity edema. Because these symptoms may be seen in other diseases, such as Baker cysts, congestive heart failure, cellulitis, thrombophlebitis, and postphlebitic syndrome, the diagnostic accuracy based on clinical presentation is poor.[16–19] Physical examination findings such as calf tenderness, unilateral calf edema, palpable calf cord, and Homan's sign, which is calf discomfort elicited by dorsiflexion, are inaccurate.[16] Less than one third of symptomatic patients who have a DVT exhibit Homan's sign.[20] In addition, one half of patients who have Homan's sign do not have a DVT.[20]

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D-dimer assays are used frequently an adjunct to the clinical assessment of patients suspected of having a DVT. Fibrinolysis of thrombin-mediated fibrin clots by plasmin produces D-dimer and fragment E. The presence of these fragments indicates that thrombogenesis has occurred. D-dimer fragments can be detected by ELISA, by latex-agglutination assay, and by whole-blood agglutination. In patients who have a low clinical suspicion for a DVT and a negative D-dimer test, DVT can be excluded safely.[21] A positive D-dimer test is nonspecific and may be caused by low levels of fibrin found in conditions of infection, inflammation, vasculitis, pregnancy, trauma, or surgery (**Box 1**).[22,23]

In the presence of a positive D-dimer or strong clinical suspicion for a DVT or PE, further evaluation with imaging studies such as ultrasound or a chest CT angiography is warranted.

PATHOPHYSIOLOGY

Virchow's triad, stasis, vessel injury, and hypercoagulability, is mentioned frequently as the three main factors that contribute to the formation of a thrombus. Hemostasis is achieved by the activation of platelets and coagulation proteins, either through the intrinsic or extrinsic pathways. The intrinsic pathway is triggered by the contact of blood with subendothelial connective tissue, activating factor XII, factor XI, prekallikrein, and high-molecular-weight kininogen and ultimately forming factor X, thrombin, and fibrin. The extrinsic pathway also leads to the formation of factor X, thrombin, and fibrin but is much faster and is thought to augment the intrinsic pathway. It involves the interaction of factor VII with tissue factor, which is found predominantly in the subendothelial tissue and cells surrounding the blood vessels.

The formation of arterial thrombus typically begins with damage to the endothelial wall, which allows platelets to adhere. Platelet glycoproteins bind to the subendothelial ligand, von Willebrand's factor, which subsequently activates the coagulation cascade, as described previously. Thrombin forms and enables fibrin deposition. The thrombus, therefore, consists of a core of platelets at the site of endothelial injury surrounded by a fibrin mesh containing trapped erythrocytes and leukocytes.

The initiation of venous thrombosis is not as well understood. In the absence of direct traumatic injury to the vessel, one model suggests that impaired venous flow results in local endothelial hypoxia and subsequent injury to the vessel wall.[9,24] Investigators have proposed that activated endothelial cells bind to tissue factor microvesicles, a form of circulating tissue factor arising from monocytes and macrophages and not bound to extravascular tissue.[24,25] Tissue factor microvesicles bind to platelets and activate the extrinsic pathway through interactions with factor VII, resulting in thrombin, fibrin, and thrombus formation.[24] Although the molecular details of venous thrombosis are still being investigated, this model suggests that the initiation of venous thrombosis is not identical to arterial thrombosis but ultimately triggers the same coagulation cascades.

Once a venous thrombus forms, a process of reorganization occurs. Following the first week of an acute DVT, clefts begin to form between the thrombus and the wall of the vein.[26–29] At this point the risk for a PE is the greatest.[30] The clot adheres to the wall of the vein; then the remodeling process ensues.[29] Capillaries extend from the vein wall, followed by fibroblasts and histiocytes that infiltrate the thrombus.[31,32] The thrombus eventually is lysed and recanalized, which occurs predominantly during the second to third weeks following thrombus formation. Fibroblasts reorganize the thrombus with fibroelastin, resulting in a thickened intima.[31,33]

Box 1
Causes of a positive D-Dimer test

- Thrombogenesis
- Infection
- Inflammation
- Vasculitis
- Pregnancy
- Trauma
- Surgery

LOCATION OF LOWER EXTREMITY DEEP VEIN THROMBOSIS AND RELATED COMPLICATIONS

Most DVTs form in the calf veins.[34,35] Postmortem and other studies have demonstrated that venous thrombi are more prone to forming adjacent to valves and venous valve pockets, larger numbers of which are found in the lower extremity.[36–39] Although the incidence of significant or fatal PEs (approximately 13%–25%) is less than that of DVTs of more proximal veins, approximately 20% to 30% of calf vein venous thrombi extend proximally.[40–42] As a venous thrombus ascends into the common femoral vein, the risk for a PE increases.[43–45] If a deep venous thrombus within the common femoral vein is left untreated, approximately 50% of patients develop an

embolism within 3 months.[43] The incidences of PE resulting from DVT of the common femoral vein, iliac vein, or inferior vena cava are approximately identical.[46]

DVT isolated to the thigh and pelvic veins, without involvement of the calf veins, also may occur. Prior studies reported a 1% to 4% incidence of DVTs isolated to the iliac veins.[47,48] With the advent of contrast enhanced CT and MR imaging that can detect venous thrombosis within the pelvic veins more readily, iliac vein thrombosis has been found to be more common than previously thought. In a study of 769 patients, Spritzer and colleagues[49] reported an acute DVT isolated to the iliac vein or inferior vena cava in 20% of the patients in their study. Eighteen percent of the patients in this study had a DVT involving the femoral as well as the iliac veins.[49]

In addition to PEs, an additional common complication of lower extremity DVTs is postphlebitic syndrome, which occurs in up to one half of patients who have DVTs.[13,50–52] Inflammation and scarring of venous valves lead to valve incompetence and reflux, resulting in venous congestion, decreased muscle perfusion, and increased tissue permeability.[13,53] Patients who have postphlebitic syndrome experience pain, swelling, heaviness, cramps, and tingling in the affected limb.[52] Thromboprophylaxis and anticoagulation therapy to prevent the formation of new or recurrent DVT are the principle means of reducing the occurrence of this complication.

An important anatomic variant to consider in the presence of iliac vein DVT or chronic left lower extremity edema is the May-Thurner syndrome. This syndrome, first described by May and Thurner in 1956,[54] results from compression of the left common iliac vein by the right common iliac artery, which crosses over the left iliac vein at the bifurcation.[55] Patients who have this physiologic stenosis of the left common iliac vein are more prone to DVT within the left iliac and lower extremity veins.[55] Other complications include varicosities, chronic venous stasis ulcers, PE, and phlegmasia cerulea dolens (PCD).[56] Awareness of this entity may provide opportunities for patients to pursue corrective surgery and potentially prevent these complications if the diagnosis is made early.

Although PCD is a complication of May-Thurner syndrome, it also is associated with other disease processes such as malignancy, any hypercoagulable state, surgery, trauma, ulcerative colitis, and congestive heart failure. A related disorder is phlegmasia alba dolens, which is thrombosis of the deep veins of the lower extremity with sparing of the superficial and collateral veins. Phlegmasia alba dolens is associated with pregnancy, is caused by compression of the left common iliac vein against the pelvic rim, and does not result in lower extremity edema.

PCD occurs when thrombosis involves the deep, superficial, and collateral veins of the lower extremity, resulting in outflow obstruction, arterial insufficiency, massive extravascular fluid sequestration, and edema.[57–59] Thrombosis extends into the capillaries in 40% to 60% of patients who have PCD, leading to irreversible ischemia, necrosis, and gangrene. PCD is a surgical emergency, and early diagnosis by ultrasound may expedite appropriate management. Although conventional venography is considered the standard imaging modality for diagnosis, MR venography can demonstrate the proximal and distal extent of thrombosis easily. The disadvantages of MR venography include susceptibility to motion artifact and the timeliness of the examination. Despite

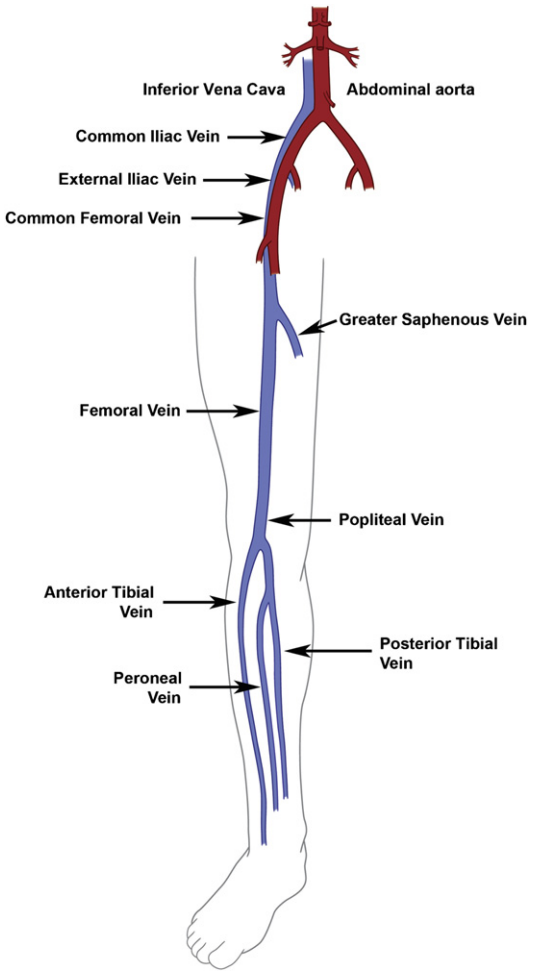


Fig. 1. The lower extremity veins. Note that the greater saphenous vein is a superficial branch.

therapy, the incidences of limb amputation and mortality remain high.

TECHNIQUE AND RELEVANT ANATOMY

Lower extremity venous ultrasound examination involves evaluation of the veins using gray-scale and color flow Doppler. The patient should be kept warm to avoid vasoconstriction and is placed in the supine position. If the veins are difficult to detect, the patient's bed may be placed in the reverse Trendelenburg position, or the patient may be asked to assume a sitting position, which has the effect of drawing blood to the dependent portions of the body.

Preferably, a higher-frequency transducer, such as a 5- to 15-MHz linear-array transducer is used. Higher frequencies allow a better spatial resolution and an improved sensitivity in detecting slow flow. Deeper veins, such as the iliac veins, may require lower frequencies such as a 3- to 5-MHz curved transducer.

Examination of the common femoral vein should begin at the junction of the femoral and profunda femoris veins (**Fig. 1**). The confluence of the greater saphenous and common femoral veins should be identified also. After color flow Doppler evaluation of the proximal common femoral vein, venous compression should be performed along the entire length of the common femoral and femoral veins at every 2 to 3 cm in the short axis with mild transducer pressure. At the level of the adductor canal, the distal femoral vein often is too deep to evaluate effectively with compression, and long-axis color flow images supplant short-axis compression.

The popliteal segment is examined with the knee slightly flexed using short-axis compression and long-axis color flow evaluation. The examination should include as much of the adductor canal as possible. The popliteal vein is superficial to the artery.

Evaluation of the three pairs of calf veins often begins at the knee and requires a variation of leg and transducer positions. The three pairs of calf veins are the anterior tibial, posterior tibial, and peroneal veins. The posterior tibial veins are located posterior to the tibia, and the peroneal veins are located deep and medial to the fibula. The posterior tibial and peroneal veins often can be imaged from the posteromedial or medial approach. The anterior tibial veins are located between the tibia and fibula and can be imaged from the anterolateral approach.

IMAGING CHARACTERISTICS OF NORMAL VEINS

The most important and basic method for evaluating veins is venous compression. This technique entails applying and releasing pressure to the

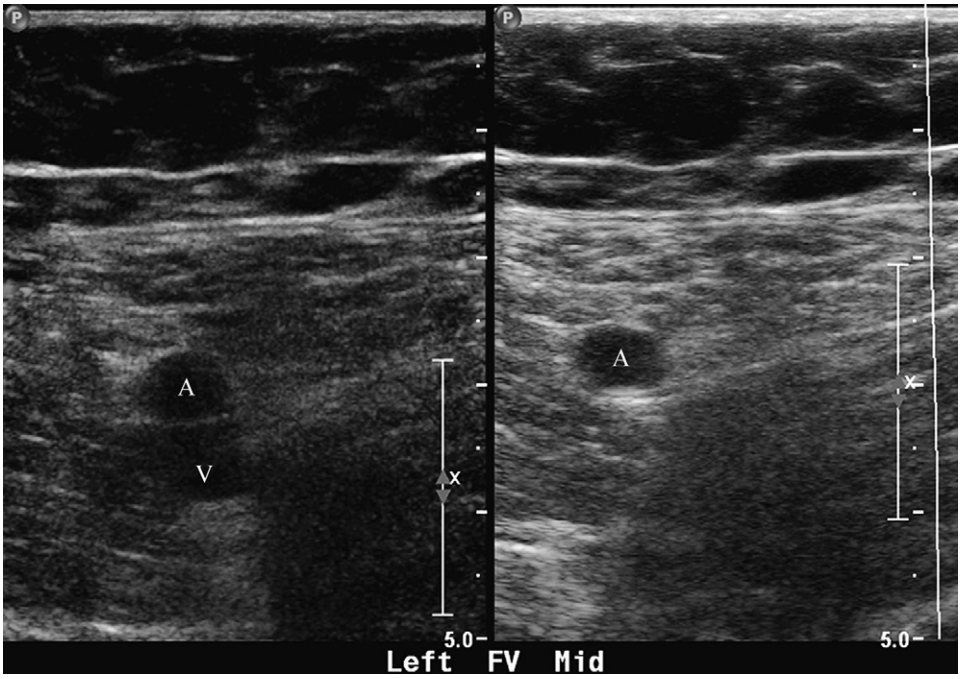


Fig. 2. Normal venous compression. Two transverse gray-scale images demonstrating the left femoral artery (A) and vein (V) without compression (*left panel*) and with compression (*right panel*). In the right panel, note complete collapse of the femoral vein with compression.

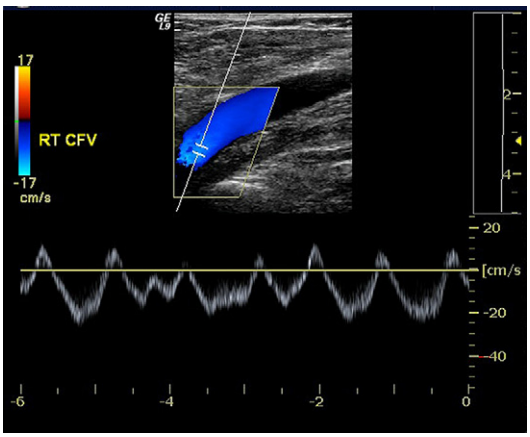


Fig. 3. Normal venous flow. Spectral Doppler tracing of the right common femoral vein demonstrating normal phasicity and spontaneous flow.

transducer every 2 to 3 cm along the entire course of the vein in the short axis. Because the vein is thin walled, it should collapse completely with compression (**Fig. 2**). A thrombus prevents the vein from collapsing.[60]

Color flow Doppler evaluation of the veins is performed in the long axis and evaluates for the presence of (1) spontaneity, (2) phasicity and direction, and (3) augmentation of flow. The presence of flow without compression indicates the presence of spontaneous flow (**Fig. 3**). Spectral Doppler analysis demonstrates the phasicity and direction of flow. Normal venous flow is respirophasic, in that the velocity of flow fluctuates according to

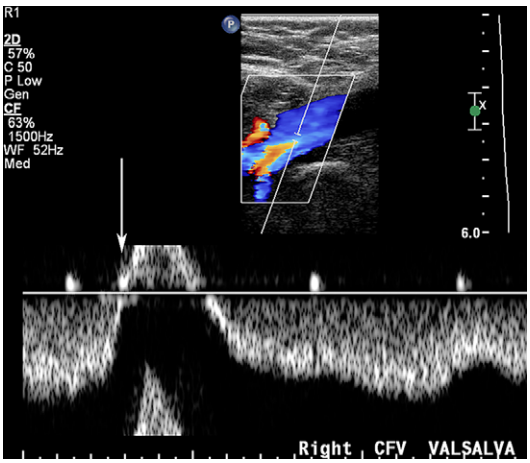


Fig. 4. Normal Valsalva response. Spectral Doppler evaluation of the right common femoral vein demonstrates cessation of antegrade flow (*arrow*) during the Valsalva maneuver. Antegrade flow resumes after the Valsalva maneuver.

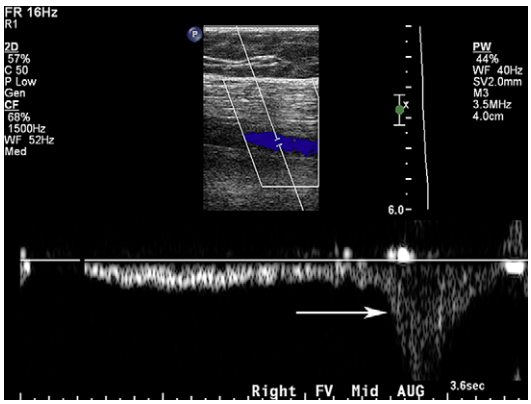


Fig. 5. Normal augmentation. Spectral Doppler tracing of the right femoral vein reveals an increase in blood flow and velocity secondary to augmentation (*arrow*).

increasing and decreasing intrathoracic pressures during normal respiration (see **Fig. 3**).

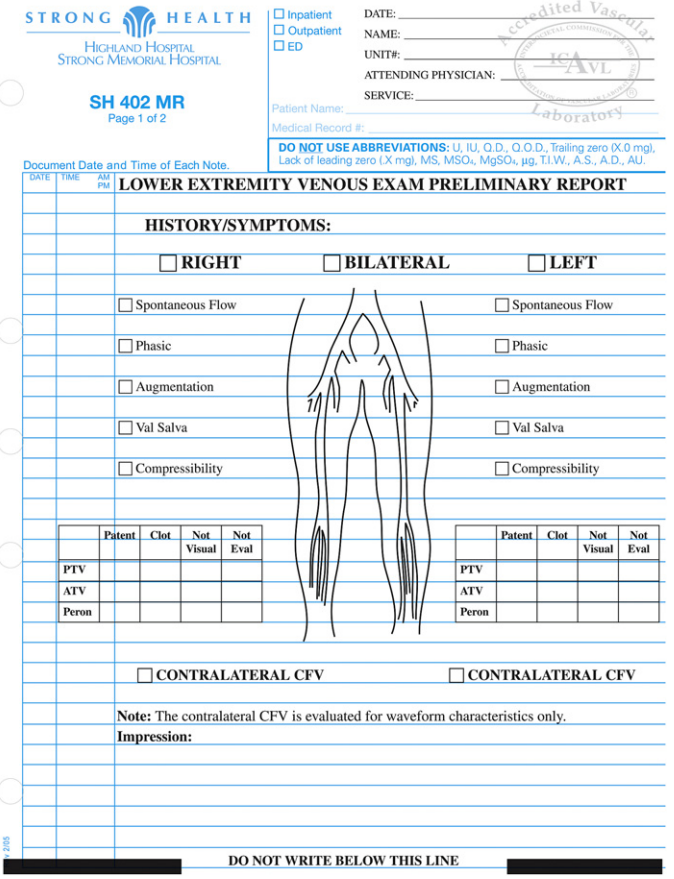
The Valsalva response is an indirect sign of thrombus proximal to the level of interrogation. This sign may detect significant venous obstruction between the thorax and the transducer. The patient is asked to take a deep inspiration and bear down (the Valsalva maneuver). If the veins are patent, the diameter of the vein should increase by 50% to 200%;[61] flow should stop completely and then resume when the patient exhales (**Fig. 4**). Flow typically is unidirectional in the presence of competent valves but occasionally may move in the retrograde direction if there are incompetent valves.

Augmentation allows detection of thrombus distal to the transducer by applying manual pressure to the calves. If the veins are patent, an increase in venous flow in the antegrade direction should be detected (**Fig. 5**).

The sonographer should document normal findings (**Box 2**) in the examination report, because they help improve sensitivity and specificity in detecting a DVT in the lower extremity veins or more proximal veins. An example from the University of Rochester School of Medicine is provided in **Fig. 6**.

Box 2
Normal imaging characteristics of a vein on ultrasound

- Compressibility
- Spontaneity of flow
- Respirophasic waveforms
- Temporary cessation of flow with Valsalva maneuver
- Increased flow with augmentation



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☐ Inpatient
☐ Outpatient
☐ ED

DATE:
NAME:
UNIT#:
ATTENDING PHYSICIAN:
SERVICE:

Patient Name:
Medical Record #:

Document Date and Time of Each Note.

DO NOT USE ABBREVIATIONS: U, IU, Q.D., Q.O.D., Trailing zero (X.0 mg), Lack of leading zero (.X mg), MS, MSO_4, $MgSO_4$, µg, T.I.W., A.S., A.D., AU.

DATE | TIME | AM PM

LOWER EXTREMITY VENOUS EXAM PRELIMINARY REPORT

HISTORY/SYMPTOMS:

☐ **RIGHT** ☐ **BILATERAL** ☐ **LEFT**

	☐ Spontaneous Flow	☐ Phasic	☐ Augmentation	☐ Val Salva	☐ Compressibility
RIGHT					
LEFT					

	Patent	Clot	Not Visual	Not Eval
PTV				
ATV				
Peron				

	Patent	Clot	Not Visual	Not Eval
PTV				
ATV				
Peron				

☐ **CONTRALATERAL CFV** ☐ **CONTRALATERAL CFV**

Note: The contralateral CFV is evaluated for waveform characteristics only.

Impression:

2/05

DO NOT WRITE BELOW THIS LINE

Fig. 6. Sample report form for a lower extremity Doppler ultrasound examination from the University of Rochester Medical Center. The report includes findings of spontaneous flow, phasicity, augmentation, Valsalva response, compressibility, and spectral Doppler evaluation of the contralateral common femoral vein (CFV). (*Courtesy of* The University of Rochester Medical Center, Rochester, NY; with permission.)

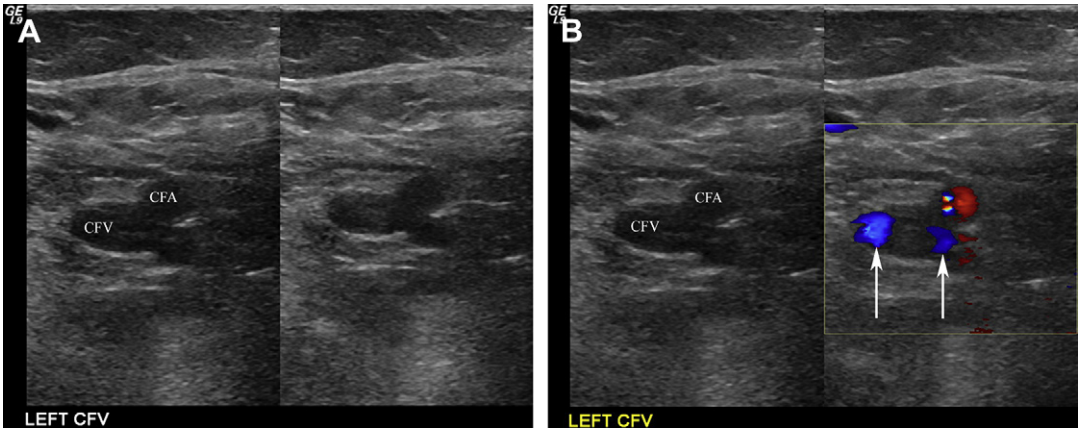


Fig. 7. Non-occlusive DVT. (*A*) Transverse gray-scale and (*B*) color flow Doppler images of the left common femoral vein without compression (*left panel*) and with compression (*right panel*) revealing persistence of the left common femoral vein lumen. In Fig. 7B, note the slightly increased echogenicity of the thrombus and partial flow (*arrows*) along the periphery of the left common femoral DVT. CFA, left common femoral artery; CFV, common femoral vein.

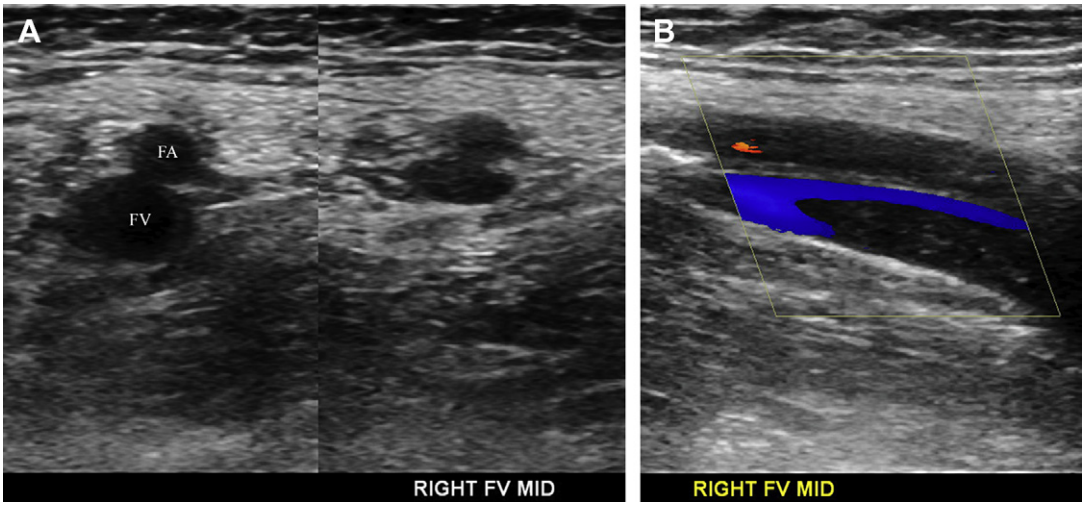


Fig. 8. Non-occlusive DVT. (*A*) Transverse gray-scale and (*B*) longitudinal color flow Doppler images of the right femoral vein revealing a non-occlusive thrombus with small amount of flow along the periphery of the thrombus. FA, right femoral artery; FV, femoral vein.

IMAGING FINDINGS OF DEEP VENOUS THROMBOSIS

Venous compression is the most reliable method of detecting an acute DVT by ultrasound. The thrombus, which prevents complete collapse of the vein, may be anechoic, hypoechoic, or hyperechoic (**Figs. 7–9**). The echogenicity of the thrombus does not depict the age of the clot accurately,[27] but anechoic and hypoechoic echotextures may suggest more recent thrombus formation. As a thrombus ages, it may become more echogenic.

Color flow Doppler demonstrates partial flow with a non-occlusive thrombus (see **Figs. 7 and 8**) and no flow with a completely occlusive thrombus (see **Fig. 9**). If a DVT is located distal to the transducer, augmentation of the calf is diminished or absent and will not result in an increase in blood flow in a more proximal segment, as is seen in normal subjects. If a DVT is located between the thorax and the transducer, the diameter of the vein will not increase during the Valsalva maneuver. This finding is nonspecific and may be seen when a more proximal vein

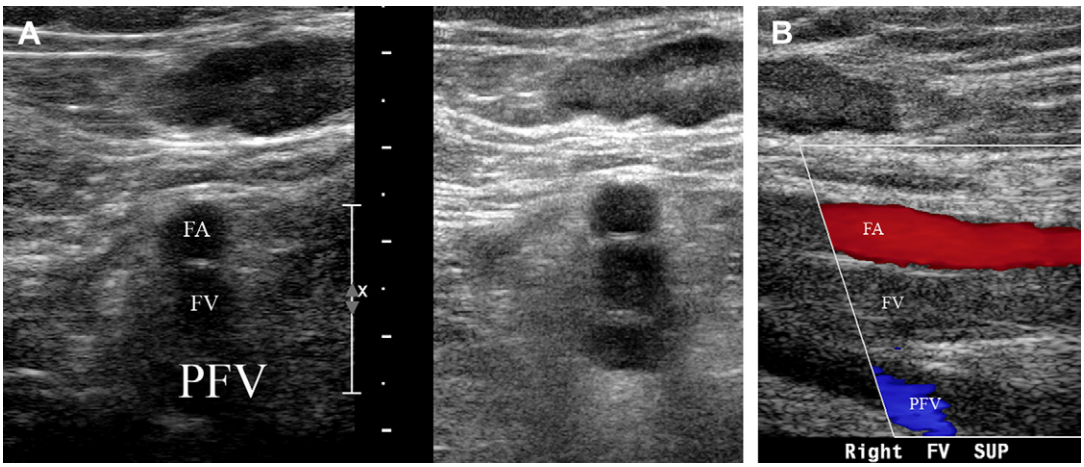


Fig. 9. Occlusive DVT. (*A*) Transverse gray-scale images of the right femoral vein without compression (*left panel*) and with compression (*right panel*) demonstrating persistence of the right femoral vein. (*B*) Longitudinal color flow Doppler images revealing lack of flow within the femoral vein. FA, right femoral artery; PFV, profunda femoris vein.

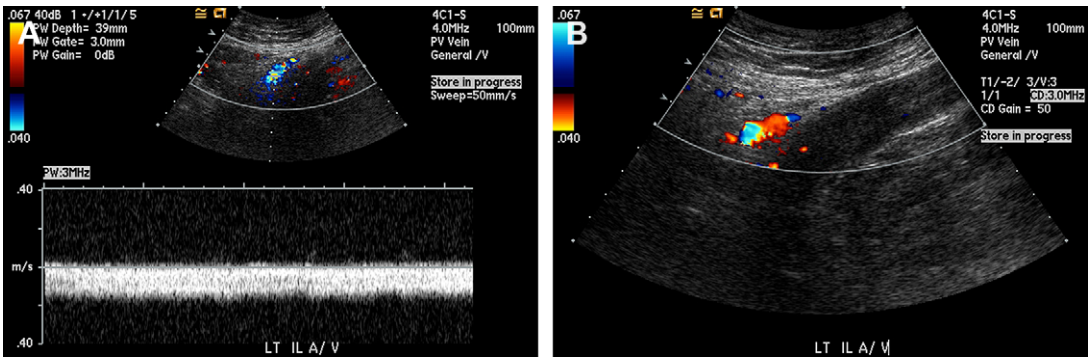


Fig. 10. Monophasic waveform. (*A*) Spectral Doppler tracing of the left external iliac vein demonstrates a monophasic waveform. (*B*) Color flow Doppler of a more proximal segment of the left external iliac vein reveals an absence of flow, representing a DVT.

is compressed by an extrinsic fluid collection or mass or in the presence of congestive heart failure. The Valsalva response also is diminished in the presence of a more proximal DVT.

Collaterals may be seen in the acute phase as well as during the chronic phase. They may be found in the immediate vicinity of a thrombosed vein or further away. Collaterals are tortuous and do not follow the course of normal vessels.

Only 20% of venous thrombi resolve with complete lysis.[62] As thrombi undergo reorganization during the healing process, many abnormalities may persist during the chronic phase, reflecting the formation of fibrous scar tissue. The vein wall may become thickened, or plaque-like foci may form along the wall giving the appearance of a thickened wall. When the venous lumen has become significantly narrowed secondary to

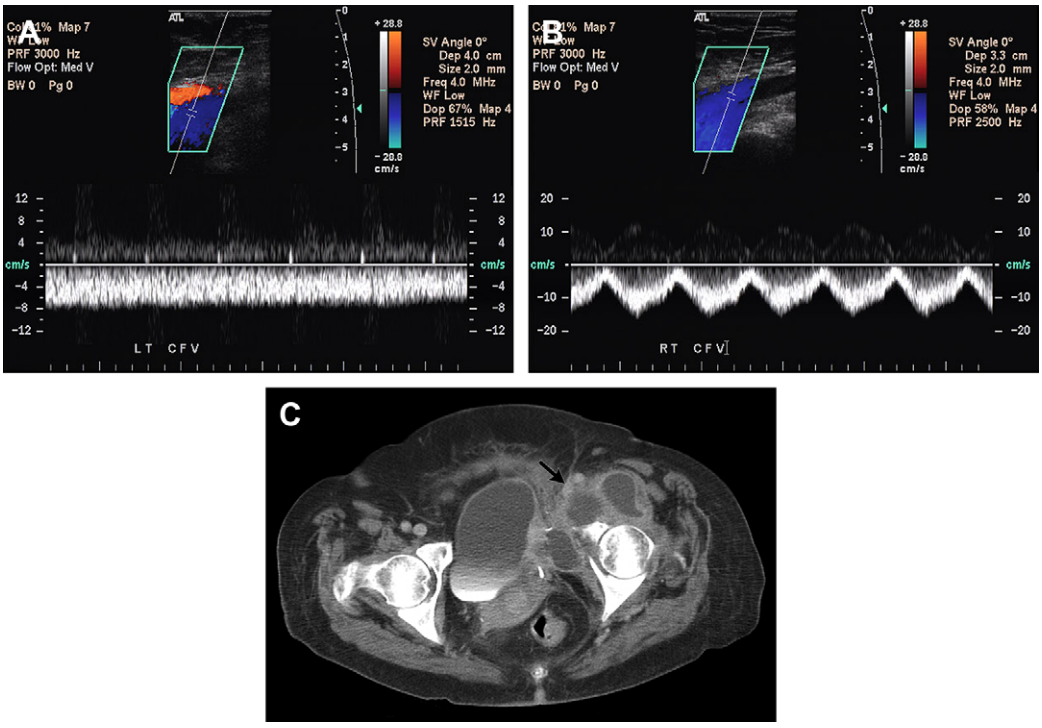


Fig. 11. Asymmetry of phasicity. Spectral Doppler waveforms of the (*A*) left and (*B*) right common femoral veins in a 71-year-old woman who has metastatic bladder carcinoma demonstrate asymmetry of waveforms, with monophasicity in the left common femoral vein. (*C*) Follow-up CT imaging after intravenous contrast exhibits large necrotic lymph nodes compressing the left external iliac vein (*arrow*).

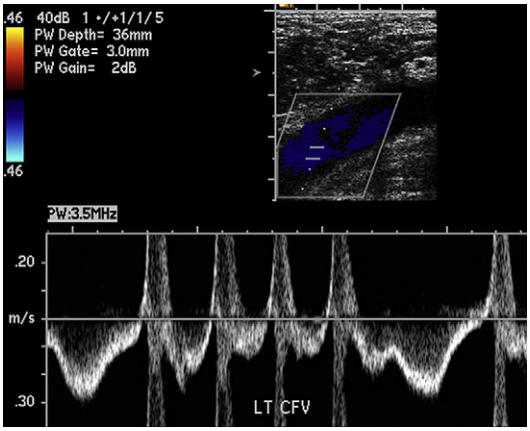


Fig. 12. Increased pulsatility. Spectral Doppler of the left common femoral vein exhibiting the increased pulsatility of flow.

Box 3
Abnormal color flow Doppler findings

- Loss of spontaneous flow
- Loss of respirophasic waveforms (monophasic or increased pulsatility)
- Diminished or absent augmentation
- Diminished or absent Valsalva response
- Collateralization

scarring, the vein may appear as a fibrous cord. Synechiae or fibrous bands also may form within the vein lumen, typically along one side of the vein. Involvement of valve cusps inevitably leads to valve dysfunction. Valve cusps become thickened and exhibit restricted movement. Venous stasis and reflux, particularly during the Valsalva maneuver, may be detected on color flow Doppler. These sequelae of DVT should not be referred to as "chronic DVT," because these findings merely reflect the healing process, not the persistence of an actual thrombus.[63] More appropriate terms are "chronic postthrombotic scarring" or "sequelae of prior DVT".[63]

Abnormal spectral analysis of the common femoral vein may include monophasic, dampened, or increased pulsatile waveforms. A monophasic waveform is the loss of normal phasicity secondary to a more proximal thrombus (**Fig. 10**), extrinsic compression of a more proximal vein by a fluid collection, mass (**Fig. 11**), or gravid uterus, or intrinsic luminal narrowing, which may be congenital or a sequela of prior thrombus or radiation therapy.[64] An asymmetric monophasic waveform of a common femoral vein, when compared with the contralateral side, indicates pathology proximal to the transducer but distal to the inferior vena cava. Increased pulsatility of the common femoral vein waveform may be seen with right heart failure (**Fig. 12**).

The sensitivity and specificity of compression ultrasound have been reported to be 95% and

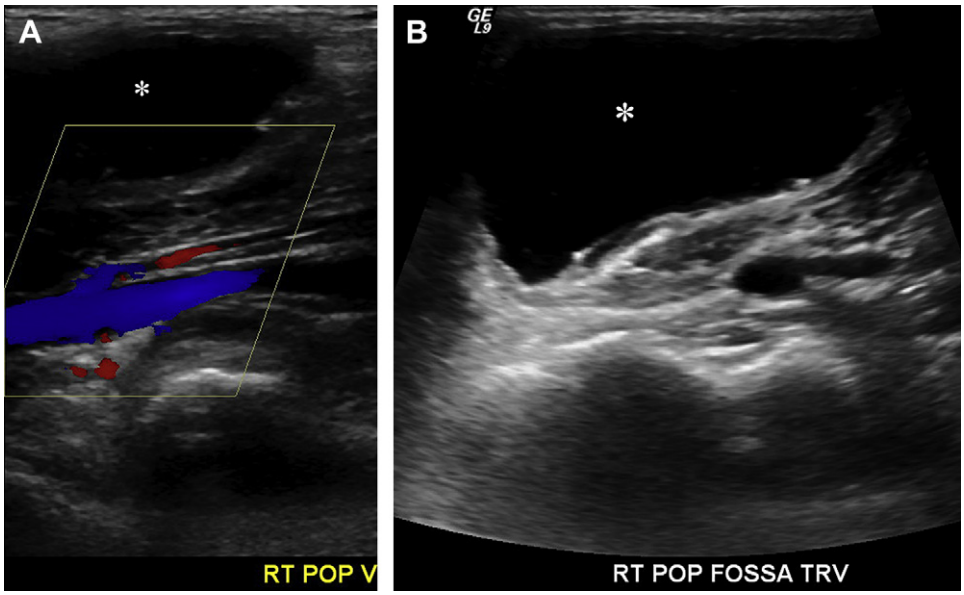


Fig. 13. (*A*) Longitudinal color flow Doppler and (*B*) transverse gray-scale images at the level of the popliteal vein revealing large hypoechoic cyst (*asterisk*) posteromedial to the vein, representing a Baker's cyst.

98%, respectively.[61] With the use of color flow Doppler ultrasound, sensitivities and specificities range from 95% to 99%.[65,66] Ultrasound findings of DVT are summarized in **Box 3**.

The differential for lower extremity swelling and pain is broad, but ultrasonography may come across findings that can explain the patient's symptoms, such as cellulitis, subcutaneous edema, and Baker cysts (**Fig. 13**), and these findings should be delineated in the impression.

SUMMARY

DVT is associated closely with PE and should be considered as part of the same disease process. Without treatment, venous thromboembolism results in a high mortality. Any clinical suspicion for a PE or DVT should prompt further evaluation. Because clinical assessment often is unreliable, a D-dimer test followed by imaging is needed to confirm or exclude a DVT or venous thromboembolic event.

Ultrasonography has become the primary imaging modality in screening for lower extremity DVT in high-risk or symptomatic patients. Other complications of DVT include postphlebitic syndrome, recurrent venous thromboembolic events, vein stenosis and scarring, and PCD.

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